UCS ALGORITHM

```
import heapq

graphs = {
    1: {
        'S': {'A': 1, 'G': 12},
        'A': {'B': 3, 'C': 2},
        'B': {'D': 3},
        'C': {'D': 1, 'G': 2},
        'D': {'G': 3},
        'G': {}
    },
    2: {
        'V1': {'V2': 9, 'V3': 4},
        'V2': {'V1': 9, 'V3': 2, 'V4': 7, 'V5': 3},
        'V3': {'V1': 4, 'V2': 2, 'V4': 1, 'V5': 6},
        'V4': {'V2': 7, 'V3': 1, 'V5': 4, 'V6': 8},
        'V5': {'V2': 3, 'V3': 6, 'V4': 4, 'V6': 2},
        'V6': {'V4': 8, 'V5': 2}
    }
}


def ucs(graph, start, goal):
    pq = [(0, start, [start])]
    visited = set()

    while pq:
        cost, node, path = heapq.heappop(pq)

        if node in visited:
            continue
```

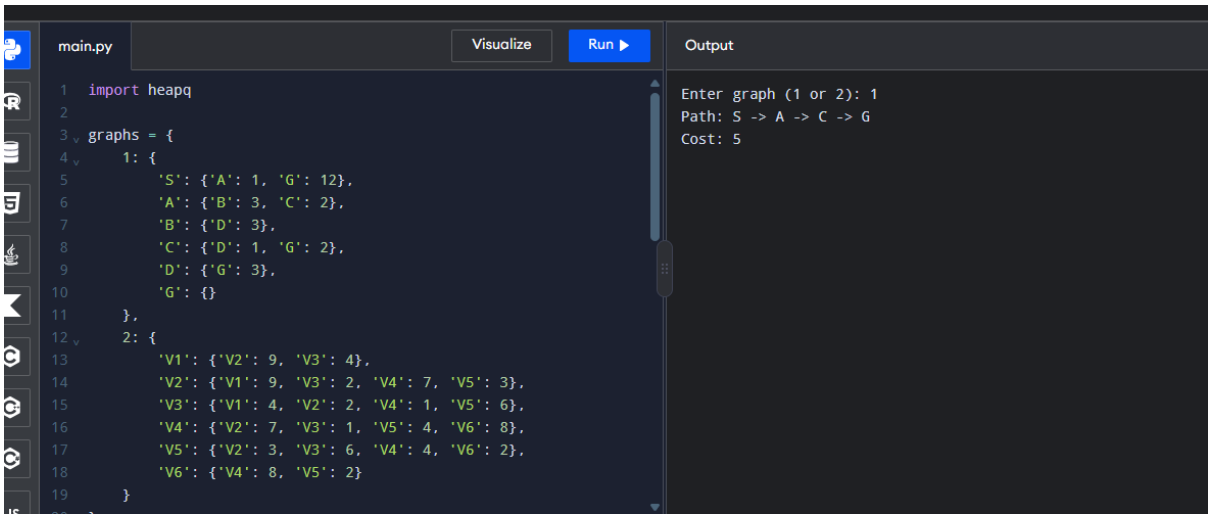
```
        visited.add(node)

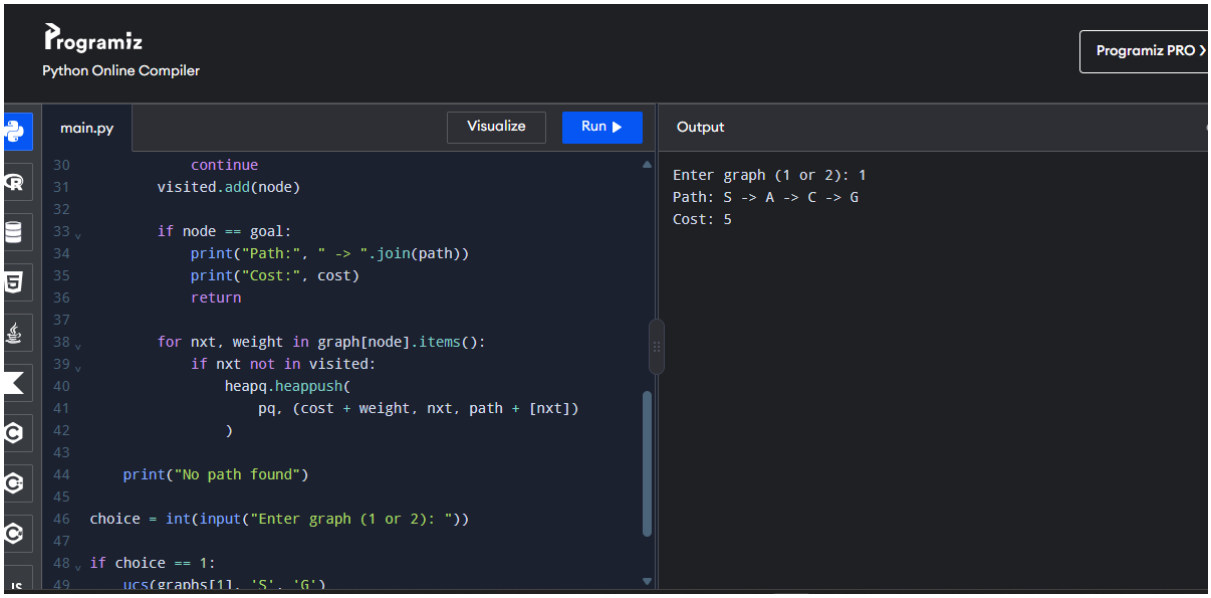
        if node == goal:
            print("Path:", " -> ".join(path))
            print("Cost:", cost)
            return

        for nxt, weight in graph[node].items():
            if nxt not in visited:
                heapq.heappush(
                    pq, (cost + weight, nxt, path + [nxt])
                )

    print("No path found")
choice = int(input("Enter graph (1 or 2): "))

if choice == 1:
    ucs(graphs[1], 'S', 'G')
elif choice == 2:
    ucs(graphs[2], 'V1', 'V6')
else:
    print("Invalid choice")
```




Programiz
Python Online Compiler
Programiz PRO
main.py
Visualize
Run
Output
continue
visited.add(node)
if node == goal:
print("Path:", " -> ".join(path))
print("Cost:", cost)
return
for nxt, weight in graph[node].items():
if nxt not in visited:
heapq.heappush(
pq, (cost + weight, nxt, path + [nxt])
)
print("No path found")
choice = int(input("Enter graph (1 or 2): "))
if choice == 1:
Enter graph (1 or 2): 1
Path: S -> A -> C -> G
Cost: 5